

2(a)(i)	<u>active transport</u> ;	1
2(a)(ii)	correct working ; e.g. $31 \times 30.5 = 945.5 \text{ then } \frac{945.5}{2870}$ or $\frac{2870}{31} = 92.58 \text{ then } \frac{30.5}{92.58}$ 32.94 / 32.9 / 33 ;	2
2(a)(iii)	thermoregulation or maintaining (constant), body / core, temperature ; I regulation	1
2(a)(iv)	substrate-linked phosphorylation ; A substrate level phosphorylation	1
2(b)	<i>rest (RQ of 0.8)</i> 1 a mix of respiratory substrates or lipids and, carbohydrates / proteins, are being respired ; <i>110 (RQ of 1.0)</i> 2 respiratory substrate used is carbohydrate / carbohydrate respired (than at rest) ; A named carbohydrate e.g. glucose <i>200 (RQ of 1.2)</i> 3 anaerobic respiration (also) occurring ; 4 reduced availability of oxygen or increased, production / release of CO₂ ;	4

Question	Answer	Marks
2(c)	any three from: 1 ref. to <u>oxygen debt</u> ; - oxygen needed to 2 convert lactate to pyruvate ; 3 convert lactate to glycogen ; 4 re-oxygenate, Hb / myoglobin ; 5 AVP ; e.g. to support a higher metabolic rate / ref. to EPOC	3